

<i>ID</i>	<i>Description</i>	<i>Type</i>	<i>Priority</i>	<i>Description</i>	<i>1</i>	<i>Status</i>
PB-01	Automated bike data collection (every 5 mins)	Technical capability	Must	Get the system to retrieve live station occupancy data from JCDecaux API every 5 minutes	Core data pipeline requirement	Completed
PB-02	Automated weather data collection (every hour)	Technical capability	Must	Get the system to retrieve weather data from OpenWeather API every hour	Required for weather display and prediction	Completed
PB-03	Store bike data locally (1-2 days)	Technical capability	Must	Store bike occupancy data that has been collected in a local database	Allows for machine learning and creation of historical charts	Completed
PB-04	Store weather data locally	Technical capability	Must	Store hourly weather data in a database for modelling and visualisation	Supports forecast capabilities and machine learning	Completed
PB-05	Design database schema	Technical capability	Must	Design and implements a database scheme for bike and weather data storage	E-R diagram required	Completed
PB-06	Display live bike stations on map	User story	Must	Show all Dublin bike stations on a map with real-time availability	Core requirement	Completed
PB-07	Colour-coded availability	User story	Must	Code stations with colours to visualise availability of bikes (green/yellow/red)	Clarity for user experience (UX)	Completed
PB-08	Station detail panel on click	User story	Must	Display details about a station when a specific marker is selected	Includes bike count information for user	Completed
PB-09	Display historical occupancy bar chart (hourly/daily)	User story	Must	Show the hourly and daily occupancy bar when a station is clicked on, using stored data	Requirement	Completed
PB-10	Display current weather and forecast	User story	Must	Show the current weather conditions and a short forecast for user	Integrate OpenWeather API data	Completed
PB-11	Route planning	User story	Should	Allow for a user to input a point of origin and destination to generate a route for them to take	Optional but desired	Completed

<i>ID</i>	<i>Description</i>	<i>Type</i>	<i>Priority</i>	<i>Description</i>	<i>1</i>	<i>Status</i>
PB-12	Predict occupancy with machine learning	Advanced feature	Must	Use the historical bike and weather data to predict the short term station availability of bikes	Higher complexity	Completed
PB-13	Filter by bike type (mechanical vs e-bike)	User story	Should	Allow for a user to filter the map stations based on the type of bike available	Depends on data breakdown available at JCDecaux API, enhances usability	Not Implemented
PB-14	Save and view favourite stations	Enhancement	Could	Allow user to mark stations as favourites and allow for quick access to saved stations	Enhances the repeat usability	Completed
PB-15	UI capability for mobile devices	Infrastructure	Could	Ensure application layout can be used across devices such as mobiles or tablets	Can be implemented using CSS and layout design, no additional backend complexity	Not Implemented
PB-16	User login	Optional feature	Optional	One of the additional features that can be selected	Optional	Completed
PB-17	Chatbot	Optional feature	Optional	One of the additional features that can be selected	Optional	Completed